

1 Consolidated Strategic Priority Matrices

The three strategic priority clusters are each assessed across four analytical dimensions per phase: (1) R&I Maturity, (2) Policy/Regulatory/Standards, (3) Operational Deployment & Adoption, and (4) Market Uptake & Strategic Sovereignty. Phases are presented as rows; the four dimensions are presented as columns. Gate rows between phases state the minimum verifiable conditions required to declare a transition. Sub-priority labels in parentheses (*e.g.*, P1, P3) identify where a specific constituent priority divergences from the cluster-level descriptor within a given cell.

The three clusters and their constituent sub-priorities are:

- **Cluster 1 — Sovereign Trusted Infrastructure:** P1 (Hardware Roots of Trust), P2 (PQC Agility), P3 (Lightweight Crypto & PhySec), P8 (Confidential Computing), P9 (Resilient-by-Design / Island Mode).
- **Cluster 2 — Cognitive Security & Autonomous Cyber Defence:** P6 (Cognitive SecOps / LLMSecOps), P10 (Proactive APT Management), AI-as-target dimension (CL1.2, CL1.3, CL4.7).
- **Cluster 3 — Federated Operational Sovereignty:** P4 (Security Interoperability via Simpl), P5 (Supply Chain Verification), P7 (Pan-European NaaS Federation).

Table 1: **Cluster 1: Sovereign Trusted Infrastructure** (P1, P2, P3, P8, P9). Transition matrix across the four methodology dimensions. Sub-priority labels in parentheses indicate where individual priorities diverge from the cluster-level descriptor within a cell. ODP scale: Applied R&D → Fragmented Piloting → Standardised Integration → Ubiquitous Operation. SSL scale: High Foreign Dependency → Emerging EU Ecosystem → Full Sovereign Autonomy.

Phase / SSL	R&I MaturityODP indicator per sub-priority	Policy, Regulatory & Standardsenforcement · certification · standardisation	Operational Deployment & Adoptionactor types · scale · sector coverage	Market Uptake & Strategic SovereigntySSL indicator · dependency reduction · industrial capacity
Phase 1 <i>Foundation & Compliance</i> <i>2025–2030</i> SSL: High Foreign Dependency	Cluster ODP: Applied R&D / Fragmented Piloting The cluster is split at baseline. (<i>P1, P9</i>) reach Fragmented Piloting: RISC-V secure element projects targeting EUCC ‘High’ are funded under Horizon Europe and EU Chips Act; Island Mode soft-state protocols are validated in 5G research testbeds. (<i>P2, P3, P8</i>) remain at Applied R&D: NIST FIPS 203/204/205 algorithms are being optimised for 8/16-bit microcontrollers under severe energy and memory constraints; Keystone and OpenTitan prototypes are under EUCC pre-evaluation; RF fingerprinting Physical-Layer Security is validated in controlled 5G lab environments; Fully Homomorphic Encryption approaches practical latency bounds in research settings only. <i>The cluster ODP is Applied R&D, driven by the majority of sub-priorities. Deployment claims for P2, P3, P8 must be grounded in named Horizon Europe or ECCC project outcomes, not technology previews.</i>	Regulatory environment: enforcing, certification nascent NIS2 Article 21(2)(e) cryptographic policies for essential entities entered full enforcement in October 2024. CRA Articles 10 and 11 (vulnerability handling and security update obligations) begin enforcement in 2027, creating immediate procurement-level demand for hardware security certification. AI Act Articles 9 and 15 establish hardware integrity requirements for high-risk AI systems and enter applicability in 2025–2026. ENISA PQC migration guidelines are published; EUCC evaluation processes are initiated for the first RISC-V prototypes, but no EU-specific EUCC ‘High’ certified sovereign secure element exists yet. CEN/CENELEC JTC 13 standards for hardware security MIMs are in draft phase. <i>The regulatory framework is enforcing existing obligations but lacks EU-specific hardware certification outputs; the critical gap is the absence of a ratified EUCC ‘High’ EU-origin product.</i>	Deployment: research consortia and national agencies only RISC-V secure element pilots are operated by 3–5 Horizon Europe consortia and national cybersecurity agencies in controlled environments. Intel TDX, AMD SEV-SNP, and ARM TrustZone dominate all operational TEE deployments across EU cloud and critical infrastructure. PQC is confined to cloud-side software implementations in pilot contexts; far-edge IoT devices run classical cryptography exclusively. Edge nodes operate with hard cloud dependencies and no validated autonomous fallback architecture meeting DORA or NIS2 business continuity thresholds. SMEs have no access to EU-sovereign hardware alternatives at viable cost. <i>Operational deployment at Phase 1 is institutional and research-concentrated. The gap between pilot existence and operational availability for regulated-sector procurement must be explicitly acknowledged.</i>	SSL: High Foreign Dependency Non-EU TPMs (Infineon SLB9670, NXP SE050, STMicroelectronics ST33) hold ≥90% of EU market share in hardware security modules. Intel, AMD, and ARM control the entire operational TEE market. No EU-designed and EU-manufactured secure element is commercially available. EU Chips Act fabrication investment for security-critical silicon is initiated but pre-production. US NIST algorithm supply chain is the sole PQC standardisation reference. Structural vulnerability is direct: geopolitical restriction of access to any of the three major TPM suppliers would immediately impair EU critical infrastructure attestation capability. <i>Quantified dependency at Phase 1 baseline: ≥90% foreign market share, zero EU-certified sovereign hardware alternatives. Sovereignty progress is measured from this specific figure.</i>
<i>Gate 1 → 2:</i>	R&I trigger: At least one EU-designed RISC-V secure element prototype has been independently validated against EUCC ‘High’ evaluation criteria and has entered the formal ENISA/national scheme certification process. Policy trigger: CRA Articles 10 and 11 are in active enforcement with documented non-compliance decisions, confirming that market incentives for certified EU hardware alternatives are financially material. Operational trigger: Documented operational deployment of hybrid ML-KEM (FIPS 203) plus ECDH-P384 PQC schemes in at least one EU critical infrastructure sector (energy or financial) at production scale, not pilot scale. Market trigger: EU Chips Act security-critical silicon fabrication pilot line has produced and delivered its first commercially viable batch of EU-origin secure elements, confirmed by independent yield and cost assessment.			

Continued...

Phase / SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Phase 2 <i>Applied & Industrial R&D alongside Federation & Standardisation 2030–2035</i> SSL: Emerging EU Ecosystem	Cluster ODP: Standardised Integration / continued Applied R&D <i>(P1, P9)</i> reach Standardised Integration: the first EUCC ‘High’ certified EU-manufactured RISC-V secure elements are in production and being integrated into critical infrastructure; automated remote attestation at fleet scale is validated; 72-hour Island Mode achieves standardised certification under 3GPP/ETSI fallback API profiles. <i>(P2, P3)</i> transition from Applied R&D to Fragmented Piloting; hybrid ML-KEM plus ECDH-P384 schemes are validated at Industrial IoT scale; lightweight PQC for sub-100 μJ zero-energy nodes is demonstrated under 6G ISAC testbeds. <i>(P8)</i> reaches Fragmented Piloting; RISC-V Keystone TEEs are operational in multi-provider Data Spaces; FHE performance approaches the threshold for selective production use in high-value financial analytics. <i>Phase 2 R&I is not purely consolidation: three of five sub-priorities are still in active applied and industrial R&D. Describing this phase as “standardisation only” misrepresents the maturation arc.</i>	EUCC certification and PQC standards formalised The EUCC ‘High’ certification of the first EU-manufactured RISC-V secure element is the definitive Phase 2 policy gate: it transforms applied research into a market-accessible, legally certifiable product and unlocks the procurement pathway for EU critical infrastructure operators. NIST FIPS 203/204/205 are adopted into ETSI TS profiles for EU deployment contexts. ENISA publishes a mandatory PQC migration timeline for critical infrastructure operators, referenced in NIS2 sector-specific implementing acts. EU Chips Act fabrication investments are active and measurable. CEN/CENELEC hardware security standards for EU-origin secure elements enter ratification. <i>The single Phase 2 policy gate is EUCC ‘High’ certification of a EU-origin product. All other policy milestones are enabling conditions, not gate conditions.</i>	Critical infrastructure and regulated sectors deploying EU alternatives EU EUCC ‘High’ certified secure elements are deployed in energy sector substations and telecommunications critical infrastructure in at least five Member States. Hybrid PQC schemes are standard in new Industrial IoT and financial sector (DORA-compliant) deployments. RISC-V Keystone TEEs operate in federated multi-provider Data Spaces alongside existing Intel TDX infrastructure. Validated 72-hour Island Mode is deployed in at least three EU 5G cross-border corridor pilots under SNS JU governance. SME-accessible PQC tooling and SBOM compliance support are available commercially. <i>Phase 2 deployment evidence requires named sector and Member State scope; anonymous “multi-sector deployment” claims do not constitute verifiable operational maturity.</i>	SSL: Emerging EU Ecosystem EU-designed secure elements achieve 15–30% market share in new critical infrastructure procurement, rising from a near-zero baseline. EU Chips Act fabrication investments have created a domestic production capacity that is commercially credible, if not yet dominant. Foreign TPMs remain the standard for legacy deployments but are no longer the only option for new regulated-sector procurement. EU open-hardware TEE ecosystem (Keystone, OpenTitan) holds commercially certifiable products. The conditions for Full Sovereign Autonomy are structurally in place — fabrication capacity, certification infrastructure, standards leadership — but have not yet reached continental operational scale. <i>SSL transition requires industrial substance, not research demonstration. 15–30% new critical infra market share is the quantified Phase 2 SSL threshold.</i>
<i>Gate 2 → 3:</i>	R&I trigger: Production-ready PQC-native RISC-V silicon is available from EU-domiciled fabrication at commercially viable yields, confirmed by independent cost-per-unit assessment against equivalent non-EU alternatives. Policy trigger: EUCC ‘High’ certification is mandated as a procurement requirement for all new EU critical infrastructure edge node deployments, with no waiver provisions for new procurements (legacy replacement programmes are time-bound). Operational trigger: Validated 72-hour Island Mode is deployed as a standard operational baseline — not a pilot — in $\geq 50\%$ of new EU NIS2-essential entity edge node procurements in the energy and telecommunications sectors. Market trigger: EU Chips Act sovereign fabrication capacity achieves $\geq 50\%$ self-sufficiency for security-critical silicon in new EU critical infrastructure procurement, measured by share of new orders fulfilled by EU-origin fabrication.			

Continued...

Phase / SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Phase 3 Autonomy & Leadership 2035–2040 SSL: Full Sovereign Autonomy	Cluster ODP: Ubiquitous Operation All five sub-priorities operate at Ubiquitous Operation. EU-designed RISC-V secure elements are the hardware default across all new EU ICT products. Fully Homomorphic Encryption is operationalised for end-to-end confidential processing from IoT to HPC, not confined to high-value analytics. Zero-energy PhySec authentication via RF fingerprinting operates as a standard device identity mechanism for battery-less mMTC nodes. Bio-inspired swarm protocols enable autonomous cyber-physical recovery without cloud intervention. Phase 3 R&I is oriented toward the next-generation frontier: neuromorphic secure compute, post-second-generation quantum hardware, and post-CMOS security architectures — EU research groups define the agenda rather than adopting it. Ubiquitous Operation requires deployment statistics, not deployment plans. Phase 3 evidence must cite procurement share data and operational audit findings, not programme milestone completions.	EU as international hardware security standards reference EUCC ‘High’ certification is mandatory for all new EU ICT products without waiver provisions. PQC-native silicon is the EU procurement default. EU hardware security standards are referenced in ISO/IEC international standardisation and in bilateral digital trade agreements. Continuous automated EUCC lifecycle maintenance replaces point-in-time audit models completely. The EU sets the terms of hardware security compliance for market access, transforming its internal market standard into a de facto international reference. Phase 3 policy maturity requires evidence of international standards influence: at least one EU hardware security standard adopted or formally referenced in an ISO/IEC or ITU-T international standard.	Security-by-Design the unconditional hardware and crypto default ≥60% of EU critical infrastructure edge nodes deploy EUCC ‘High’ certified EU-designed secure elements. FHE is operational end-to-end from IoT sensor to HPC compute node. Autonomous Island Mode provides validated 72-hour survivability as a mandatory operational baseline across all NIS2-essential and DORA-critical edge deployments. No new EU ICT product enters the market without a hardware root of trust. The question “does this deployment have hardware-level integrity?” has no non-trivial answer for any new EU deployment. Phase 3 deployment normalisation: ≥60% market penetration for EU secure elements in critical infra; 100% of new EU ICT products with hardware RoT as a default.	SSL: Full Sovereign Autonomy EU-designed and EU-manufactured secure elements hold ≥60% of new critical infrastructure procurement market share. Foreign legacy dependency operates under binding replacement programmes with defined timelines. EU sovereign RISC-V silicon and open-hardware TEE architectures are exported to and adopted by partner jurisdictions. EU PQC and hardware TEE standards are the international reference. Structural exposure to geopolitical disruption of any single non-EU hardware supplier has been eliminated for new deployments. The EU has moved from defensive sovereignty (reducing dependency) to offensive strategic posture (defining the terms of global hardware security competition). Full Sovereign Autonomy is verified by ≥60% market share in new critical infra procurement plus the existence of binding replacement programmes for legacy foreign hardware — not by 100% market share.

Table 2: **Cluster 2: Cognitive Security & Autonomous Cyber Defence** (P6, P10, AI-as-target dimension from CL1.2, CL1.3, CL4.7). Transition matrix across the four methodology dimensions. The AI-as-target dimension (model integrity, adversarial robustness, algorithmic accountability) is labelled (*AI-T*) where it diverges from P6 or P10 descriptors.

Phase / SSL	R&I MaturityODP indicator per sub-priority	Policy, Regulatory & Standardsenforcement · certification · standardisation	Operational Deployment & Adoptionactor types · scale · sector coverage	Market Uptake & Strategic SovereigntySSL indicator · dependency reduction · industrial capacity
Phase 1 <i>Foundation & Compliance 2025–2030</i> SSL: High Reliance on US Foundation Models	Cluster ODP: Applied R&D / Siloed AI Copilots (<i>P6</i>) reaches Siloed AI Copilots: commercial LLMs (GPT-4-class) assist human SOC analysts in alert triage, threat report generation, and MITRE ATT&CK technique mapping; no EU-sovereign LLM for security operations exists at production scale; NWDAF (3GPP TS 23.288) telemetry pipelines are integrated in 5G SA pilot deployments for anomaly detection. (<i>P10</i>) is at Multi-Layered Alert Correlation: ML-based UEBA is applied to APT IoC detection in research and pilot contexts; APT dwell time reduction from the 150–200 day average is demonstrated in controlled settings. (<i>AI-T</i>) is at Applied R&D: adversarial robustness testing (prompt injection, model evasion, DataBOM verification) is an academic research function without operational certification frameworks. <i>Phase 1 R&I distinguishes three sub-dimensions with different maturity levels. Conflating them into a single ODP cell obscures that AI-as-target capability is the least mature and the most consequential gap.</i>	AI Act enforcing; conformity assessment infrastructure absent AI Act Articles 9 (risk management), 13 (transparency), and 15 (adversarial robustness) enter applicability in 2025–2026, establishing mandatory requirements for autonomous security AI as high-risk systems. However, no EU conformity assessment bodies with capacity to certify autonomous security agents are operational at the start of Phase 1. ETSI GS ZSM 009 (closed-loop automation) is published but not mandated. NIS2 Article 21(2)(b) incident handling procedures are being operationalised across essential entities. ENISA AI cybersecurity threat landscape guidelines are published. The critical regulatory gap is the absence of a standardised adversarial robustness test suite applicable to security-specific AI agents. <i>The AI Act creates obligations before the conformity assessment infrastructure exists to fulfil them. This 24–36 month gap is the dominant Phase 1 policy risk for this cluster.</i>	Deployment: SOC copilot use by large enterprises and national CSIRTs US-platform LLMs (GPT-4-class, under data processor agreements) operate as analyst copilots in large enterprise and national CSIRT SOC. Rule-based SIEMs (Splunk/Cisco, IBM QRadar) remain the primary detection infrastructure across EU regulated sectors. NWDAF-integrated AI analytics operate in 3–5 5G SA deployments run by major EU telecom operators. No certified autonomous security containment agent is in operational deployment anywhere in the EU. APT dwell times of 150–200 days in EU enterprise environments reflect the gap between AI-assisted detection capability and operational deployment at scale. SMEs lack access to AI-powered SOC tooling at viable cost. <i>Phase 1 deployment is concentrated in the largest institutional actors (national CSIRTs, Tier-1 operators, major financial entities). The gap between these deployments and SME access defines the adoption challenge for Phase 2.</i>	SSL: High Reliance on US Foundation Models The EU SOC tooling market is dominated by US-domiciled platforms: Splunk (Cisco), IBM QRadar, and Microsoft Sentinel collectively control the EU enterprise SIEM market. The foundation models used for SOC copilot functions are US-origin (OpenAI, Anthropic, Google) and are accessed under data processor agreements, creating persistent extra-territorial data exposure. No EU-sovereign foundation model at production scale for security operations exists. EuroHPC compute capacity is insufficient for frontier-class security model training. The adversarial robustness validation market is pre-commercial. <i>Quantified dependency: US platforms hold ≥80% of EU enterprise SIEM market; US-origin models are the sole source for LLM-based SOC tooling. This specific dependency structure is what Phases 2 and 3 must reduce.</i>
Gate 1 → 2:	R&I trigger: At least one EU-sovereign Small Language Model (≤10B parameters) trained on GDPR-compliant security corpora has been validated against an independently benchmarked EU critical infrastructure SOC use case at production-equivalent scale. Policy trigger: AI Act conformity assessment bodies with capacity to certify autonomous security AI as high-risk systems under Articles 9 and 15 are operational and have issued at least one certificate to a security AI agent product. Operational trigger: At least one EU-sovereign LLMSecOps agent is in certified operational deployment in a named NIS2-essential entity, demonstrating APT dwell time below 72 hours in a documented production environment. Market trigger: A standardised adversarial robustness test suite for security AI agents, developed under CEN/CENELEC or ETSI TC CYBER governance, has been published and is referenced in at least one EU procurement framework for critical infrastructure AI tools.			

Continued...

Phase / SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Phase 2 <i>Applied & Industrial R&D alongside Federation & Standardisation 2030–2035</i> SSL: Sovereign EU Edge LLMs Emerging	Cluster ODP: Federated Edge Agents / Automated Triage <i>(P6)</i> reaches Federated Edge Agents: EU-sovereign SLMs (≤10B parameters) trained on MITRE ATT&CK-aligned corpora are operational at the edge for real-time SOC analytics without cloud inference dependency; Guardian nodes execute autonomous Zero-Trust reconfigurations per NIST SP 800-207. <i>(P10)</i> reaches Automated Triage and LLMSecOps Containment: LLMSecOps agents ingest CTI feeds, track APT lateral movement against behavioural frameworks, and orchestrate incident triage autonomously under ETSI GS ZSM 009 governance; APT dwell time ≤72 hours in certified pilot deployments. <i>(AI-T)</i> reaches Standardised Integration: adversarial robustness validation pipelines (prompt injection, model evasion, backdoor detection) are standardised and mandatory under AI Act Article 9 conformity assessment; continuous RLHF retraining and circuit-breaker mechanisms are in production. <i>Phase 2 R&I includes active applied and industrial R&D for all three sub-dimensions — swarm AI protocols, post-quantum AI security, and next-generation adversarial testing frameworks — not merely consolidation of Phase 1 results.</i>	AI Act conformity assessment operational: the Phase 2 gate The single Phase 2 policy gate is the operationalisation of a standardised AI Act conformity assessment framework for autonomous security agents, enabling certified deployment without per-case regulatory approval. ETSI GS ZSM 009 is mandated within EU NaaS federation governance frameworks. A CEN/CENELEC-standardised adversarial robustness test suite for security AI is published. MITRE ATT&CK behavioural framework is formally referenced in NIS2 sector-specific incident handling guidance. EuroHPC AI Factory compute contracts for EU-sovereign security model training are active. <i>Without operational AI Act conformity assessment infrastructure, autonomous security agents cannot be certified for regulated-sector deployment regardless of technical readiness. This is the dominant Phase 2 policy dependency.</i>	Certified EU sovereign AI agents in critical infrastructure SOC EU SLM-based LLMSecOps agents are in certified operational deployment in energy, financial (DORA), and telecommunications critical infrastructure SOC in ≥3 Member States. Guardian nodes execute autonomous Zero-Trust reconfigurations under human-in-the-loop escalation protocols for Tier-1 critical actions. NWDAF-integrated analytics are mandatory in 5G SA deployments. Deception infrastructure (honeynets, credential canaries, fake API endpoints per CL2.6) is deployed alongside AI detection systems, providing high-fidelity ground truth for model calibration. APT dwell time ≤72 hours demonstrated in pilot critical infrastructure deployments. <i>Phase 2 deployment evidence requires named critical infrastructure sector, Member State scope, and a documented APT dwell time metric from an operational — not simulated — environment.</i>	SSL: Sovereign EU Edge LLMs Emerging EU-sovereign SLMs cover ≥30% of critical infrastructure SOC deployments in regulated sectors, rising from zero at baseline. EuroHPC AI Factories supply training infrastructure for EU-origin security models, reducing dependence on US hyperscaler compute. US-platform SIEMs retain market share in legacy deployments but face procurement competition from certified EU alternatives. Adversarial robustness validation is a commercially available service with EU-domiciled providers. EU AI security tool exports to partner jurisdictions have begun, indicating the transition from security consumer to security producer posture. <i>SSL evidence at Phase 2: ≥30% critical infra SOC market share for EU-sovereign AI tools; at least one documented EU AI security tool export event to a non-EU partner jurisdiction.</i>
<i>Gate 2 → 3:</i>	R&I trigger: EU-certified swarm AI coordination protocols enabling decentralised autonomous threat intelligence synthesis have passed independent security evaluation and are deployed in at least one cross-border EU critical infrastructure corridor under ETSI GS ZSM 009 governance. Policy trigger: EU AI security standards (adversarial robustness profiles, autonomous agent accountability frameworks) are adopted or formally referenced in at least one ISO/IEC or ITU-T international standard, confirming EU regulatory leadership beyond domestic jurisdiction. Operational trigger: APT dwell time ≤24 hours is documented in a named NIS2-essential entity operational deployment (not a simulation), attributable to autonomous AI-driven detection and containment. Market trigger: EU-sovereign AI security tools hold ≥50% market share in new critical infrastructure SOC deployments across at least five Member States, displacing US-platform SIEMs as the procurement default in regulated sectors.			

Phase / SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Phase 3 Autonomy & Leadership 2035–2040 SSL: Global Leadership in Trustworthy AI Security	Cluster ODP: Ubiquitous AI Immune System All three sub-dimensions reach Ubiquitous Operation. Swarm-based agentic AI defends the EU Cloud-Edge Continuum via self-evolving, decentralised threat intelligence without any central coordination dependency. Moving Target Defense continuously shuffles IP addressing, network topology, and software diversity; validated APT reconnaissance success rates fall below 5% in certified operational environments. Adversarial robustness validation is continuous and fully automated: no security AI agent operates in EU critical infrastructure without an active conformity certification. Phase 3 R&I addresses post-swarm frontiers: quantum-native AI security architectures, brain-inspired neuromorphic threat detection, and AI-to-AI adversarial dynamics beyond current threat models. Phase 3 ODP evidence requires operational statistics from production deployments: APT dwell time figures, MTD effectiveness metrics, and autonomous containment accuracy rates from certified environments — not from cyber range simulations.	EU AI security governance as international reference EU AI Act frameworks for trustworthy autonomous security agents are adopted internationally and referenced in bilateral digital partnership agreements. Autonomous agent certification operates at continental scale without per-case regulatory review. Zero-touch network management under ETSI GS ZSM 009 governs all EU NaaS operations. The EU AI Office and ENISA jointly define the international standard for security AI accountability. EU AI security governance has transitioned from reactive adaptation to agenda-setting: other jurisdictions adopt EU frameworks to access EU market and partnership programmes. Phase 3 policy leadership is verified by international adoption events — not by the number of EU instruments in force. At least one non-EU jurisdiction formally adopting EU security AI governance standards constitutes the Phase 3 policy evidence threshold.	Autonomous continental-scale AI cyber defence as operational default APT dwell time ≤24 hours in all certified EU NIS2-essential critical infrastructure deployments. MTD is deployed as a mandatory operational baseline. Swarm AI coordination operates without central cloud dependency: each EU Cloud-Edge node participates in a self-organising threat intelligence mesh. Deception infrastructure is fully integrated into the autonomous response loop, providing continuous model calibration ground truth. Human intervention is reserved for exceptional legal escalation scenarios; routine security operations are fully autonomous within the governance framework. The Phase 3 deployment test: is autonomous AI defence the unconditional operational baseline, or is it still a premium programme? 100% of new critical infrastructure deployments incorporating autonomous AI security constitutes the evidence threshold.	SSL: Global Leadership in Trustworthy AI Security EU-sovereign AI security tools hold ≥50% of new critical infrastructure SOC deployments across the EU. US-platform SIEMs are legacy infrastructure under active replacement programmes with defined timelines. EU AI security models are exported to and commercially adopted by partner jurisdictions. EU sets the commercial terms for security AI market access: certification against EU AI Act Article 9/15 profiles is a de facto global procurement requirement. The EU has transitioned from defensive sovereignty to offensive strategic posture: it defines the technical terms on which the global security AI industry must compete. Full sovereign leadership in this cluster requires both domestic dominance (≥50% critical infra SOC market share) and international influence (at least two partner jurisdictions formally adopting EU security AI governance). Both conditions must be met.

Table 3: **Cluster 3: Federated Operational Sovereignty** (P4, P5, P7). Transition matrix across the four methodology dimensions. P4 (Simpl Interoperability), P5 (Supply Chain Verification), and P7 (NaaS Federation) are labelled where their maturity trajectories diverge within a cell.

Phase / SSL	R&I MaturityODP indicator per sub-priority	Policy, Regulatory & Standardsenforcement · certification · standardisation	Operational Deployment & Adoptionactor types · scale · sector coverage	Market Uptake & Strategic SovereigntySSL indicator · dependency reduction · industrial capacity
Phase 1 <i>Foundation & Compliance 2025–2030</i> SSL: Fragmented; Hyperscaler Dominance (P7); Emerging EU Ecosystem (P4)	Cluster ODP: Fragmented Piloting (P4, P7) / Disclosure Phase (P5) <i>(P4)</i> is at Fragmented Piloting: Simpl security MIM pilots are launched in three Common European Data Spaces (Manufacturing, Health, Energy) with manual policy coordination and proof-of-concept implementations. <i>(P7)</i> is at Bilateral Interconnection: harmonised CAMARA Network APIs are deployed in 2–3 bilateral 5G corridor agreements between individual EU telecom operators; cross-border edge compute roaming does not exist at operational scale. <i>(P5)</i> is at Supply Chain Disclosure: CRA-mandated static SBOM declarations are being submitted as point-in-time compliance artefacts without runtime verification; Digital Product Passport pilot programmes are initiated under the EU Ecodesign Regulation; Zero-Knowledge Proof verification for supply chain attestation is at Applied R&D stage. <i>The three sub-priorities have meaningfully different Phase 1 maturity levels. P5 is the least mature and the most consequential shared dependency: Runtime SBOM verification is a gate condition for both P4 (MIM compliance trust) and Cluster 1 (TEE integrity claims).</i>	Multiple legislative obligations activating; standards nascent CRA SBOM obligations (Annex I Part II) enter enforcement in 2027, mandating static SBOM disclosure for all products with digital elements entering the EU market. NIS2 Article 21(2)(h) makes supply chain risk assessment mandatory for essential entities. DORA Article 28 requires continuous ICT third-party risk monitoring for financial sector entities. Data Act interoperability obligations activate, creating procurement-level pressure for cross-provider security interface standardisation. The Simpl-Open official launch with initial MIM definitions constitutes the Phase 1 technical policy trigger for P4. IPCEI-CIS enters its deployment phase. GSMA CAMARA API Alliance standardisation work is active but not yet mandated in EU regulatory instruments. <i>Phase 1 policy activity is high-volume but fragmented: multiple legislative instruments activate simultaneously without a unified interoperability mandate that would create coherent market pull for Simpl-based standardisation.</i>	Deployment: bilateral pilots and Data Space proof-of-concepts Three Common European Data Space security pilots operate Simpl-based MIM proof-of-concepts with manual policy coordination; cross-provider security policy enforcement is not automated. CAMARA API deployments are confined to 2–3 bilateral 5G corridor agreements between single operator pairs. Static SBOM declarations are submitted as CRA compliance artefacts but are not verified at runtime; supply chain risk assessment is vendor-declared rather than independently attested. Hyperscalers (AWS, Azure, GCP) retain ≈65% of EU cloud market through proprietary API ecosystems. Cross-border AI model migration requires bespoke bilateral legal review per transfer event. SME-accessible SBOM tooling and MIM compliance support are not yet standardised or commercially affordable. <i>Phase 1 deployment is bilateral and pilot-scale. The gap between proof-of-concept MIM implementations and automated enforcement across the majority of Data Spaces defines the core operational challenge for Phase 2.</i>	SSL: Mixed — Emerging EU Ecosystem (P4) / Hyperscaler Dominance (P7) <i>(P4)</i> has reached Emerging EU Ecosystem: the Simpl open-source stack exists and is being piloted with EU institutional backing. <i>(P7)</i> remains at Hyperscaler Platform Dominance: AWS, Azure, and GCP collectively control ≈65% of EU cloud market; no EU NaaS federation operates at commercially relevant scale; cross-border edge compute is fragmented into national silos. <i>(P5)</i> is at High Reliance on Fragmented Non-EU Auditing: supply chain verification is vendor-declared under US-dominated certification ecosystems (Common Criteria national schemes, SOC2). The cluster’s dominant SSL profile is foreign dependency, driven by the weight of P7 market structure. <i>P4 and P7 require separate SSL tracking in Phase 1: conflating them into a single cluster SSL figure obscures that Simpl has achieved Emerging Ecosystem while NaaS federation has not.</i>

Continued...

Phase / SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Gate 1 → 2:	R&I trigger: At least one Common European Data Space has deployed automated (not manually coordinated) Simpl MIM-based cross-provider security policy enforcement, with documented evidence that enforcement is cryptographically verifiable rather than declaratively asserted. Policy trigger: Simpl security MIM definitions are published in candidate CEN/CENELEC standard form and referenced as a compliance pathway in at least one EU sector-specific NIS2 or Data Act implementing act. Operational trigger: CAMARA-based cross-border edge compute roaming is operational between at least three distinct EU telecom operator pairs (not a single bilateral agreement), with documented SLA performance and compliance evidence. Market trigger: Runtime SBOM tooling with cryptographic execution proofs is available as a commercially supported product or service from at least two EU-domiciled vendors, establishing the market infrastructure for Phase 2 mandatory adoption.			
Phase 2 <i>Applied & Industrial R&D alongside Federation & Standardisation 2030–2035</i> SSL: Growing Pan-EU Sovereignty	Cluster ODP: Standardised Integration (P4) / Federated Edge Roaming (P7) / Dynamic Runtime Verification (P5) <i>(P4)</i> reaches Standardised Integration: automated policy translation via standardised Simpl MIMs is deployed across the majority of active Common European Data Spaces; federated identity attestation and cross-provider compliance reporting are automated. <i>(P7)</i> reaches Federated Edge Roaming: CAMARA-based cross-border roaming operates across ≥5 EU operator pairs; “Data Visa” mechanisms enable dynamic AI model migration without manual legal review in certified corridors. <i>(P5)</i> transitions from disclosure to dynamic runtime verification: Runtime SBOMs with cryptographic proof-of-execution verify containerised components as they migrate across multi-provider nodes; ZKP batch verification of supply chain attestation is operationally validated; Digital Product Passports are mandatory under the EU Ecodesign Regulation. <i>Phase 2 R&I activity includes industrial R&D for ZKP real-time verification (currently batch-only due to 10–100× overhead), semantic security ontology standardisation, and legal engineering of the “Schengen for Cloud” framework — not solely standardisation of existing prototypes.</i>	Security MIMs as CEN/CENELEC standards: the Phase 2 gate The single Phase 2 policy gate is the adoption of Simpl security MIMs as ratified CEN/CENELEC standards, referenced as mandatory compliance requirements in the EU Cloud Rulebook. This transforms MIM adoption from voluntary best-practice to market-access obligation for providers serving EU-regulated sectors. DPP becomes mandatory under the EU Ecodesign Regulation 2024 for ICT hardware categories. Data Act full enforcement creates binding procurement-level interoperability obligations. DORA continuous ICT third-party monitoring is fully operationalised in the financial sector. IPCEI-CIS cross-border NaaS governance framework is operational. <i>The Phase 2 policy gate is the CEN/CENELEC ratification of security MIMs. Voluntary Simpl adoption without this standards event does not constitute Phase 2 policy maturity.</i>	Cross-sector rollout; SME participation; Runtime SBOM in production Simpl MIMs are deployed across the majority of active Common European Data Spaces with automated cross-provider enforcement. Runtime SBOMs with cryptographic execution proofs verify supply chain components in critical infrastructure deployments in at least five Member States. CAMARA cross-border roaming operates across ≥5 EU operator pairs with documented SLA performance. “Data Visa” mechanisms are operational in certified cross-border AI model migration corridors. SME-accessible Simpl MIM compliance tooling and Runtime SBOM services are commercially available from EU-domiciled providers. Unverified supply chain components are automatically quarantined in isolated network slices. <i>Phase 2 deployment must show geographic breadth (≥5 Member States), cross-sector presence (≥3 Data Spaces under automated MIM enforcement), and SME participation in at least one deployment context. Single-sector rollout does not constitute Phase 2 operational maturity.</i>	SSL: Growing Pan-EU Sovereignty EU Data Space infrastructure operating under Simpl achieves ≥30% non-hyperscaler share in new regulated-sector cloud procurement. Runtime SBOM ecosystem is commercially mature with multiple competing EU-domiciled vendors. EU NaaS consortium (under IPCEI-CIS) operates cross-border compute at pilot commercial scale, demonstrating that EU providers can federate to compete with hyperscaler geographic coverage. DPP traceability reduces procurement risk for EU critical infrastructure buyers by making supply chain provenance verifiable. Foreign hyperscaler market share in EU regulated sectors declines for the first time from its ≈65% Phase 1 baseline. <i>Phase 2 SSL evidence: ≥30% non-hyperscaler share in new regulated procurement; first documented market share decline for non-EU hyperscalers in EU regulated cloud procurement.</i>

Continued...

Phase / SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Gate 2 → 3:	R&I trigger: ZKP verification for supply chain attestation has achieved real-time per-transaction performance (eliminating the current batch-only constraint) in at least one production critical infrastructure deployment, confirmed by independent performance benchmarking. Policy trigger: A “Schengen for Cloud” legal framework — or equivalent EU legal instrument resolving US CLOUD Act extra-territorial jurisdiction conflict for federated EU-sovereign NaaS workloads — has been enacted, providing the jurisdictional basis for single-market operation of federated compute. Operational trigger: The EU NaaS federation (under IPCEI-CIS or its successor) operates at commercial scale across ≥ 10 EU telecom operator members, offering cross-border edge compute services as a standard commercial product with documented customer deployments outside its founding consortium. Market trigger: EU supply chain governance frameworks (Runtime SBOM plus ZKP attestation) are adopted or formally referenced by at least two non-EU partner jurisdictions as part of digital trade or partnership agreements, confirming that EU governance leadership is internationally exportable.			
Phase 3 Autonomy & Leadership 2035–2040 SSL: Full Federated Sovereign Autonomy	Cluster ODP: Ubiquitous Operation across all sub-priorities All three sub-priorities reach Ubiquitous Operation. Simpl-based security is the universal standard for cross-provider Data Space operation in the EU; no alternative interoperability model holds market relevance in regulated sectors. ZKP-verified supply chain attestation operates at real-time transaction scale for all critical infrastructure component verification. The EU NaaS federation operates as a single commercial entity legally equivalent to a unified jurisdiction for data workloads. Phase 3 R&I pursues next-generation frontiers: post-ZKP cryptographic verification, semantic AI-to-AI policy negotiation, and legal engineering of the next-generation “Digital Single Market” framework beyond cloud-edge compute. Phase 3 ODP evidence: Simpl MIM compliance rate in new regulated EU cloud procurement; ZKP verification throughput in production critical infrastructure deployments; EU NaaS federation revenue relative to hyperscaler EU revenue in regulated sectors.	EU federated governance as international model The “Schengen for Cloud” legal framework is enacted and operational, resolving US CLOUD Act jurisdiction conflicts for federated EU-sovereign workloads and enabling single-market legal operation of cross-border compute. EU Cloud Rulebook is mandatory for all new cloud deployments in EU regulated sectors without exception. EU supply chain governance standards (Runtime SBOM plus ZKP) are referenced in international supply chain security standards (ISO/IEC) and in digital trade agreements with partner jurisdictions. The EU has transitioned from rule-taker to rule-setter in international cloud and supply chain governance. Phase 3 governance leadership is verified by two events: enactment of the “Schengen for Cloud” framework (domestic) and formal adoption of EU supply chain governance standards by ≥ 2 non-EU partner jurisdictions (international).	Provider-agnostic EU trust fabric as continental operational default A single provider-agnostic EU trust fabric operates continuously across all active Common European Data Spaces with real-time automated security orchestration and zero manual policy coordination. ZKP-verified supply chain attestation eliminates all unverified components from EU critical infrastructure deployments in regulated sectors. The EU NaaS federation is a standard commercial service operating at hyperscaler-comparable scale, with EU operators collectively competitive on cost and coverage. Security-by-Design is not an add-on: every new cloud-edge deployment in the EU incorporates Simpl MIM compliance, Runtime SBOM verification, and cross-border sovereignty assurance as architectural defaults. Phase 3 deployment normalisation: Simpl MIM compliance in $\geq 90\%$ of new regulated EU cloud deployments; ZKP supply chain verification in 100% of new critical infrastructure component procurement; EU NaaS federation operating as a single commercial entity with documented non-founding-member customers.	SSL: Full Federated Sovereign Autonomy Non-EU hyperscaler market share in EU regulated cloud procurement falls below 40% for new deployments (from $\approx 65\%$ at baseline), with a binding trajectory toward continued reduction. EU NaaS federation achieves hyperscaler-comparable geographic coverage and commercial viability across the EU. Supply chain governance internationally adopted: at least two partner jurisdictions have formally adopted EU Runtime SBOM plus ZKP attestation standards as domestic procurement requirements. The EU has removed the structural dependency that allowed US CLOUD Act jurisdiction to apply to EU-regulated data: federated sovereignty is legally guaranteed, not only technically implemented. Full Federated Sovereign Autonomy requires three concurrent conditions: $< 40\%$ hyperscaler new-procurement share; “Schengen for Cloud” legally enacted; ≥ 2 partner jurisdictions formally adopting EU supply chain governance standards. Achieving any two without the third does not constitute this SSL level.

2 Progression Matrix: v2 Format

Table 4: Progression Matrix — Priority Cluster 1: Sovereign Trusted Infrastructure (P1: Hardware RoT, P2: PQC Agility, P3: Lightweight Crypto & PhySec, P8: Confidential Computing, P9: Island Mode). Each phase is assessed across four analytical dimensions per the generalised methodology framework. Critical Enabler rows state the single named gate condition controlling the transition into the following phase.

Priority Cluster 1 — Sovereign Trusted Infrastructure Sub-priorities: P1, P2, P3, P8, P9			
Analytical Dimension	Phase 1: Foundation & Compliance (2025–2030)	Phase 2: Applied & Industrial R&D alongside Federation & Standardisation (2030–2035)	Phase 3: Autonomous Leadership & Frontier Innovation (2035–2040)
(1) R&I Maturity <i>ODP: Applied R&D [cluster composite; see sub-priority decomposition below]</i>	RISC-V secure element prototypes validated against EUCC evaluation criteria under EU Chips Act pilot funding; NIST FIPS 203/204/205 algorithms optimised for 8/16-bit microcontrollers within 32 KB RAM constraints, per NIST IR 8413 performance benchmarks <i>[ref]</i> ; RF fingerprinting CSI/CFO device authentication validated in controlled mMTC environments; RISC-V Keystone TEE attestation protocols tested against heterogeneous edge fleet configurations <i>[ref]</i> ; soft-state Island Mode credential caching validated in 5G disconnection simulation environments under SNS JU governance <i>[ref]</i> . The cluster ODP is Applied R&D — the lowest common denominator driven by P2, P3, and P8; P1 and P9 have reached Fragmented Piloting and are not undeployed.	EU-manufactured RISC-V secure elements complete EUCC ‘High’ evaluation and enter industrial production; hardware-optimised ML-KEM and ML-DSA validated on EU silicon for constrained-device targets; RISC-V Keystone TEEs deployed in multi-provider EU Data Spaces with continuous remote attestation <i>[ref]</i> ; FHE performance benchmarks achieve operational viability for defined AI pipeline workloads; Island Mode swarm coordination protocols validated in multi-operator cross-border corridor deployments under IPCEI-CIS oversight. Advanced applied R&D continues in parallel on zero-energy PhySec, neuromorphic secure compute, and FHE generalisation — confirming that Phase 2 is not a purely standardisation phase but a dual-track applied and industrial R&D stage.	EU research groups set the international frontier agenda in lightweight PQC, confidential computing, and physical-layer security without adopting externally defined research directions. Second-generation PQC hardware co-designed with EU secure elements; neuromorphic secure edge compute integrated with zero-energy PhySec; quantum-native TEE architectures for post-CMOS environments; bio-inspired Island Mode swarm protocols achieving autonomous cyber-physical recovery at continental scale without any cloud connectivity or central coordination.
	<i>Transition criterion: at least one functional prototype per sub-priority validated against the minimum EU compliance baseline (NIS2, CRA, or AI Act as applicable).</i>	<i>Transition criterion: production-ready EUCC-evaluated reference implementation for P1; validated performance benchmarks on EU silicon for P2, P3, P8; 72-hour Island Mode survivability demonstrated operationally for P9.</i>	<i>Transition criterion: the capability is included as a mandatory baseline requirement in at least one EU-wide standard or procurement framework with no waiver provisions for new deployments.</i>

Continued. . .

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
(2) Policy, Regulatory & Standards <i>Status: Partially enforced / Standards under development</i>	CRA Articles 10 and 11 full enforcement from October 2027 establishes the mandatory market condition for hardware security certification investment <i>[ref]</i>. NIS2 Article 21(2)(e) cryptographic policy obligations active from October 2024 <i>[ref]</i>. NIST FIPS 203/204/205 finalized August 2024 providing algorithm standard basis <i>[ref]</i>; ENISA PQC transition guidelines in publication; EUCC evaluation methodology available but no EU-manufactured secure element has completed High assurance evaluation; 3GPP TS 33.501 Release 18 provides 5G security architecture for Island Mode integration <i>[ref]</i>. DORA Article 9 ICT resilience testing requirements apply to P9 <i>[ref]</i>; AI Act Article 9 risk management obligations apply to TEE-attested autonomous workloads <i>[ref]</i>. <i>Critical policy gap: no EU-specific ratified standard for RISC-V secure element security evaluation existed at the time of writing; this must be addressed in Phase 1 CEN/CENELEC JTC 13 work programme.</i>	EUCC ‘High’ certification of the first EU-manufactured RISC-V secure element issued — the single most critical policy gate condition for this cluster. CEN/CENELEC hardware security standards for RISC-V secure elements ratified; ETSI TS 103 744 hybrid key exchange profiles adopted as mandatory profiles in EU critical infrastructure procurement <i>[ref]</i>; EUCS ‘High+’ certification operational for TEE-backed cloud services <i>[ref]</i>; 3GPP Release 19/20 specifies PhySec and ISAC security requirements for 6G non-terrestrial network integration aligned with 3GPP TR 33.836 <i>[ref]</i>. <i>Critical policy gate: EUCC ‘High’ certification of the first EU-manufactured RISC-V secure element is the binding gate condition controlling the Phase 2 SSL transition for P1, P2, P3, and P8 simultaneously.</i>	EUCC ‘High’ mandatory baseline for all new EU ICT products in regulated sectors; EU PQC profiles referenced in ISO/IEC international standards, establishing EU as a standards-setting jurisdiction rather than an adopter; continuous automated EUCS/EUCC certification operational at continental scale without manual audit cycles; AI Act Article 9 conformity assessment for hardware-backed autonomous security AI fully operational and referenced internationally <i>[ref]</i>; DORA Article 11 recovery time objectives achieved autonomously by Island Mode architecture without programmatic enforcement <i>[ref]</i>. <i>End-state governance criterion: at least one EU standard in this cluster’s capability domain adopted as or formally referenced in an ISO/IEC or ITU-T international standard.</i>

Continued...

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
(3) Operational Deployment & Adoption <i>Deployment: Research consortia and institutional pilots</i>	Intel TDX and AMD SEV-SNP dominate all production TEE deployments in EU cloud infrastructure; RISC-V secure elements at prototype stage in 3–5 EU-funded research consortia; edge nodes universally operate with hard cloud dependencies and no validated autonomous fallback in production; hybrid PQC pilots operate in cloud research environments but not in IoT field deployments; Island Mode soft-state protocols validated in 5G research corridors under SNS JU programme management [ref]. Deployment actors are limited to large institutional early adopters (national cybersecurity agencies, telecom operators in EU corridor pilots, EU-funded research consortia); SME adoption is absent due to tooling immaturity. <i>Transition criterion: documented operational deployment across at least two EU critical infrastructure sectors or Member State jurisdictions with published findings.</i>	EU RISC-V secure elements deployed in energy and telecommunications critical infrastructure under IPCEI-CIS programme oversight; hybrid ML-KEM plus ECDH-P384 key encapsulation operational in Industrial IoT corridor deployments across 3+ Member States per ETSI TS 103 744 [ref]; multi-party confidential computing pipelines active in 3+ EU Data Spaces with EUCS ‘High+’ compliance verification; 72-hour Island Mode survivability validated operationally in 5G cross-border corridor deployments without WAN connectivity; SME-accessible crypto-agility OTA tooling available via Digital Europe Programme. <i>Transition criterion: deployment across five or more Member States or three or more EU critical infrastructure sectors, with documented SME participation in at least one deployment context.</i>	At least 60% of EU critical infrastructure uses EU-designed EUCC ‘High’ certified secure elements as default procurement; all new EU IoT deployments ship with FIPS 203/204/205-compliant PQC without bespoke integration overhead [ref]; FHE operationalised end-to-end for privacy-preserving AI pipelines from IoT to HPC; bio-inspired Island Mode swarms achieve DORA-mandated recovery objectives autonomously; Security-by-Design is structurally indistinguishable from product functionality across the EU ICT market. <i>End-state deployment criterion: at least 80% of new ICT deployments include this capability as a standard component without dedicated integration effort; the compliance distinction between “secure” and “standard” products ceases to be meaningful.</i>
(4) Market Uptake & Strategic Sovereignty SSL: High Foreign Dependency	Non-EU TPMs (Infineon SLB9670, NXP SE050, STMicroelectronics ST33 series) hold >90% of EU critical infrastructure secure element market under ISO/IEC 11889 [ref] and TCG TPM 2.0 specifications [ref]; EU Chips Act fabrication pilot lines initiated but first certified secure elements are 3–5 years from production; no EU-manufactured EUCC ‘High’ certified component available in any product category; PQC implementation ecosystem dominated by US/UK cryptography research outputs; Island Mode tooling absent from EU commercial product portfolios. <i>SSL baseline: High Foreign Dependency across all five sub-priorities. Market transition criterion: at least one EU-designed solution achieves a verifiable market entry event (product launch, procurement award, or EUCC/EUCS certification).</i>	SSL: Emerging EU Ecosystem EU alternatives hold 15–30% market share in new critical infrastructure secure element procurement; EU Chips Act fabrication ramping toward 50% self-sufficiency target; first commercial EU TEE products (Keystone-based [ref]) competing at product level with Intel TDX/AMD SEV-SNP; foreign dependency reduced from structural to situational in regulated-sector procurement; EU industrial actors present in sovereign hardware procurement across 5+ Member States. <i>Market transition criterion: EU-designed solutions hold at least 30% of new critical infrastructure procurement market share, or foreign dependency formally reduced below the threshold identified at baseline.</i>	SSL: Full Sovereign Autonomy EU Chips Act sovereign fabrication meets ≥60% demand for security-critical silicon across all product categories; EU TEE, secure element, and PQC solutions commercially exported to partner jurisdictions; zero structural foreign dependency in new critical infrastructure deployments; EU sets technical terms of competition in secure silicon globally through EUCC certification requirements and procurement weight; foreign legacy deployments under active replacement programmes with binding timelines. <i>End-state SSL criterion: EU-designed solutions hold at least 60% of new critical infrastructure procurement; foreign legacy under binding replacement; EU sovereignty is self-sustaining, not programme-managed.</i>

Continued. . .

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
<i>Critical Enabler</i> <i>Gate Phase 1 → Phase 2:</i>	CRA Articles 10 and 11 enforcement (October 2027) [ref] creating mandatory market demand for hardware security certification and establishing the commercial conditions that justify EU Chips Act fabrication investment and EUCC ‘High’ evaluation resource allocation.	EUCC ‘High’ certification of the first EU-manufactured RISC-V secure element [ref] constituting the gate condition that transitions the cluster from research-grade to market-grade sovereign hardware and unlocks EU Chips Act fabrication scale-up investment justification.	EU Chips Act sovereign EUV-capable fabrication achieving ≥60% market self-sufficiency in security-critical silicon removing the last structural foreign dependency in the infrastructure trust chain and enabling Full Sovereign Autonomy as a self-sustaining market condition.
<i>Sub-Priority ODP / SSL Decomposition — individual maturity tracking per constituent priority across phases. Composite cluster ODP and SSL rows above represent the lowest common denominator; this section preserves intra-cluster maturity heterogeneity.</i>			
<i>P1: Hardware Roots of Trust & EUCC ‘High’</i>	ODP: Fragmented Piloting SSL: High Foreign Dependency	ODP: Standardised Integration SSL: Emerging EU Ecosystem	ODP: Ubiquitous Operation SSL: Full Sovereign Autonomy
<i>P2: PQC Agility for Edge/IoT</i>	ODP: Applied R&D SSL: High Foreign Dependency	ODP: Fragmented Piloting SSL: Emerging EU Ecosystem	ODP: Ubiquitous Operation SSL: Full Sovereign Autonomy
<i>P3: Lightweight Crypto & Physical-Layer Security</i>	ODP: Applied R&D SSL: High Foreign Dependency	ODP: Standardised Integration SSL: Emerging EU Ecosystem	ODP: Ubiquitous Operation SSL: Full Sovereign Autonomy
<i>P8: Confidential Computing & Privacy-Preserving AI</i>	ODP: Applied R&D SSL: High Foreign Dependency	ODP: Fragmented Piloting SSL: Emerging EU Ecosystem	ODP: Ubiquitous Operation SSL: Full Sovereign Autonomy
<i>P9: Resilient-by-Design & Autonomous Island Mode</i>	ODP: Fragmented Piloting SSL: High Foreign Dependency	ODP: Standardised Integration SSL: Contained Local Survivability	ODP: Ubiquitous Operation SSL: Systemic EU Resilience

Table 5: Progression Matrix — Priority Cluster 2: Cognitive Security and Autonomous Cyber Defence (P6: LLMSecOps, P10: Proactive APT Management, AI-as-Target: CL1.2, CL1.3, CL4.7). Assessed across four analytical dimensions per the generalised methodology framework in this document.

Priority Cluster 2 — Cognitive Security and Autonomous Cyber Defence Sub-priorities: P6, P10, AI-as-Target (CL1.2, CL1.3, CL4.7)			
Analytical Dimension	Phase 1: Foundation & Compliance (2025–2030)	Phase 2: Applied & Industrial R&D alongside Federation & Standardisation (2030–2035)	Phase 3: Autonomous Leadership & Frontier Innovation (2035–2040)
(1) R&I Maturity <i>ODP: Applied R&D / Siloed AI Copilots [cluster composite]</i>	NWDAF-integrated ML-based UEBA deployed in 5G standalone pilot environments per 3GPP TS 23.288 [ref]; EU-sovereign SLMs (<10B parameters) in early training on MITRE ATT&CK-aligned corpora under GDPR-compliant data governance [ref]; AI Act Article 9 conformity assessment methodology in development for autonomous security agents [ref]; adversarial robustness testing frameworks for prompt injection and model evasion under active research; APT dwell time reduction from 150–200 day baseline targeted via ML-assisted IoC correlation in pilot enterprise environments.	Edge-native EU-sovereign SLMs certified under AI Act Article 9 conformity assessment and deployed in critical infrastructure SOC environments; LLMSecOps agents executing autonomous Zero-Trust reconfigurations per NIST SP 800-207 [ref]; Moving Target Defense (MTD) validated in cross-operator EU corridor simulation environments per ETSI GS ZSM 009 closed-loop automation [ref]; RLHF continuous correction pipelines maintaining model accuracy against evolving APT behavioural profiles; deception technology infrastructure (honeynets, credential canaries per CL2.6) providing AI training ground truth. Advanced applied R&D continues on swarm AI coordination protocols and self-evolving adversarial robustness frameworks.	Swarm-based agentic AI coordination protocols active across EU Cloud-Edge Continuum; self-evolving threat intelligence synthesis via decentralised AI without central coordination dependency; EU frontier research defines international agenda in trustworthy autonomous security AI; second-generation adversarial robustness frameworks address post-quantum AI attack surfaces; validated APT reconnaissance success rates below 5% in operational deployment environments.
	<i>Transition criterion: at least one EU-sovereign SLM trained on MITRE ATT&CK-aligned EU threat corpora with documented adversarial robustness validation; at least one NWDAF-integrated UEBA pilot with published dwell-time reduction evidence.</i>	<i>Transition criterion: EU-sovereign SLM deployed operationally in certified critical infrastructure SOC contexts with documented APT dwell-time reduction to under 72 hours; adversarial robustness validated under CEN/CENELEC-standardised framework.</i>	<i>End-state criterion: EU AI security research groups are agenda-setters, not fast-followers; autonomous APT dwell time below 24 hours as operational norm across EU critical infrastructure.</i>

Continued...

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
(2) Policy, Regulatory & Standards <i>Status: AI Act entering force; conformity assessment capacity building</i>	AI Act Articles 9, 13, and 15 enter into force and apply to autonomous security containment agents as high-risk AI systems [ref]; NIS2 Article 21(2)(b) incident handling procedure obligations create institutional demand for AI-augmented SOC capabilities [ref]; ETSI GS ZSM 009 closed-loop automation standards provide the governance framework for zero-touch security orchestration [ref]; MITRE ATT&CK framework adopted as mandatory behavioural taxonomy for EU threat intelligence sharing; EU AI Office establishing conformity assessment body capacity for high-risk AI certification — this institutional bottleneck represents the primary Phase 1 policy risk: insufficient conformity assessment capacity will delay Phase 2 certified operational deployment. <i>Critical policy gap: AI Act conformity assessment body capacity for high-risk autonomous security AI must be established in Phase 1; a 24–36 month bottleneck in Phase 2 certification is the primary systemic risk for this cluster.</i>	Standardised AI Act Article 9 conformity assessment framework for autonomous security agents operational, enabling certified deployment without per-case regulatory approval; ETSI GS ZSM 009 formally adopted in EU telco sector procurement requirements; MITRE ATT&CK-aligned pan-EU CTI sharing standards ratified under NIS2 implementing acts [ref]; CEN/CENELEC AI security agent assurance standard published; NIST SP 800-207 ZTA integration requirements adopted in EU critical infrastructure security baselines [ref]. <i>Critical policy gate: standardised AI Act conformity assessment framework for autonomous security agents is the single gate condition enabling certified Phase 3 swarm-level operational deployment.</i>	EU AI Act framework for autonomous security AI recognised internationally and referenced in bilateral digital trade agreements; EU standards for AI security agent trustworthiness adopted as ISO/IEC international reference; GDPR Article 22 HITL requirements and AI Act Article 13 explainability obligations are met by governance-by-default rather than programmatic enforcement [ref]; EU as the reference jurisdiction for responsible autonomous cyber defence AI deployment globally. <i>End-state governance criterion: at least one EU AI security standard adopted as or referenced in an ISO/IEC or ITU-T international standard; EU regulatory framework cited in at least two bilateral digital trade agreements.</i>

Continued. . .

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
(3) Operational Deployment & Adoption <i>Deployment: US-domiciled SIEM dominant; AI copilots in pilot SOCs</i>	LLMs operating as analyst copilots in EU enterprise SOC environments with human analysts retaining all containment authority; NWDAF telemetry routed to centralised AI analytics in 5G standalone deployments of 3+ EU operators; US-domiciled SIEM platforms (Splunk/Cisco, IBM QRadar) augmented with ML correlation layers; adversarial robustness validation confined to research environments with no operational certification framework in force; no EU-sovereign foundation model for security operations in production deployment; APT containment remains manual with AI-assisted alert triage only. <i>Transition criterion: documented AI-augmented SOC deployment in at least two EU critical infrastructure sectors with published dwell-time reduction evidence; at least one EU-sovereign SLM pilot in an operational security context.</i>	EU-sovereign SLMs deployed in critical infrastructure SOC environments across 5+ Member States; autonomous Zero-Trust reconfigurations operational in certified EU 5G core deployments; LLMSecOps agents ingesting CTI and tracking lateral movement in real time against MITRE ATT&CK behavioural frameworks; APT dwell time reduced to under 72 hours in EU critical infrastructure under autonomous agent operation; purple-team automation (CL1.4) conducting continuous adversarial validation against live staging environments; deception technology (CL2.6) generating high-fidelity APT ground truth for continuous agent retraining. <i>Transition criterion: EU-sovereign SLM deployed operationally in 5+ Member States; autonomous containment certified under AI Act Article 9 for at least one critical infrastructure sector.</i>	Swarm AI defends EU Cloud-Edge Continuum autonomously with APT reconnaissance success rates below 5% in operational environments; Moving Target Defense reshuffles network topology, IP addressing, and software diversity at continental scale per ETSI GS ZSM 009 zero-touch automation <i>[ref]</i>; deception infrastructure generates real-time adversarial ground truth continent-wide; autonomous agents meet AI Act Article 13 explainability requirements without human intervention for standard containment decisions; zero-touch SOC operation is the operational baseline for all EU critical infrastructure. <i>End-state deployment criterion: zero-touch security orchestration is the operational default; manual analyst intervention reserved for novel threat escalation only, not routine containment.</i>
(4) Market Uptake & Strategic Sovereignty SSL: High Reliance on US Foundation Models	US-domiciled SIEM platforms (Splunk/Cisco, IBM QRadar) dominate EU enterprise SOC deployments; EU organisations structurally dependent on non-EU threat intelligence feeds (Mandiant/Google, CrowdStrike, Microsoft Sentinel) for APT indicator attribution; no EU-sovereign LLM for security operations at production scale; EuroHPC AI Factory compute capacity insufficient for training frontier security foundation models within Phase 1 timeframe, constraining EU models to SLM/LCM scale (<10B parameters). <i>SSL baseline: High Reliance on US Foundation Models / High Dependency on Global Threat Feeds. Market transition criterion: at least one EU-sovereign SLM achieves a production deployment event in an EU critical infrastructure SOC.</i>	SSL: Sovereign EU Edge LLMs / SLMs Emerging EU-sovereign SLMs deployed in ≥30% of critical infrastructure SOC environments; US-domiciled SIEM dependency reduced to legacy deployments in regulated sectors under active replacement; EU CTI sharing infrastructure (CONCORDIA successor <i>[ref]</i>) operational as sovereign alternative to US-domiciled threat intelligence feeds; EuroHPC AI Factory compute sufficient for training EU security SLMs at operational scale; first EU AI security products exported to partner jurisdictions. <i>Market transition criterion: EU-sovereign SLMs hold at least 30% of EU critical infrastructure SOC deployments; measurable reduction in US-domiciled SIEM dependency for new procurement in regulated sectors.</i>	SSL: Global Leadership in Trustworthy AI Security EU-sovereign SLMs cover ≥50% of EU critical infrastructure SOC deployments; US-domiciled SIEM dependency eliminated for new deployments in regulated sectors; EU AI security technology exported to and adopted by partner jurisdictions as governance and technology reference; EU as global standard-setter for trustworthy autonomous cyber defence; continental APT dwell time from 150–200 day baseline reduced to under 24 hours as operational norm. <i>End-state SSL criterion: EU AI security models adopted as reference implementations by partner jurisdictions; EU is a net exporter of AI security governance standards, not an importer.</i>

Continued...

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
Critical Enabler <i>Gate Phase transition:</i>	AI Act Articles 9 and 15 enforcement (2025–2026) [ref] establishing mandatory risk management and adversarial robustness requirements for autonomous security AI, creating the regulatory framework for certified Phase 2 deployment and incentivising EU-sovereign model development as the compliance-safe alternative to US-domiciled foundation models.	Standardised AI Act conformity assessment framework for autonomous security agents enabling certified operational deployment at scale without per-case regulatory approval, removing the certification bottleneck that otherwise delays Phase 3 swarm-level operational autonomy across all three clusters.	EU-certified self-evolving AI swarm protocols achieving full operational deployment under ETSI GS ZSM 009 zero-touch automation governance [ref] with pan-EU CTI-sharing infrastructure providing sovereign adversarial intelligence at continental scale.
Sub-Priority ODP / SSL Decomposition — individual maturity tracking per constituent priority across phases.			
<i>P6: Cognitive SecOps (LLMSecOps) & Automated Threat Intelligence</i>	ODP: Siloed AI Copilots SSL: High Reliance on US Foundation Models	ODP: Federated Edge Agents SSL: Sovereign EU Edge LLMs Emerging	ODP: Ubiquitous AI Immune System SSL: Global Leadership in Trustworthy AI
<i>P10: Proactive APT Management</i>	ODP: Multi-Layered Alert Correlation & Hunting SSL: High Dependency on Global Threat Feeds	ODP: Automated Triage & LLMSecOps Containment SSL: Sovereign Pan-EU CTI & Behavioural Analytics	ODP: Ubiquitous Moving Target Defense (MTD) SSL: Full Autonomous Counter-Adversarial Resilience
<i>AI-as-Target (CL1.2, CL1.3, CL4.7): Model Integrity, Runtime SBOMs, AI Trustworthiness</i>	ODP: Applied R&D SSL: High Foreign Dependency	ODP: Standardised Integration SSL: Emerging EU Ecosystem	ODP: Ubiquitous Operation SSL: Full Sovereign Autonomy

Table 6: Progression Matrix — Priority Cluster 3: Federated Operational Sovereignty (P4: Security Interoperability via Simpl, P5: Supply Chain Verification, P7: Pan-European NaaS Federation). Assessed across four analytical dimensions per the generalised methodology framework in this document.

Priority Cluster 3 — Federated Operational Sovereignty Sub-priorities: P4, P5, P7			
Analytical Dimension	Phase 1: Foundation & Compliance (2025–2030)	Phase 2: Applied & Industrial R&D alongside Federation & Standardisation (2030–2035)	Phase 3: Autonomous Leadership & Frontier Innovation (2035–2040)
(1) R&I Maturity <i>ODP: Fragmented Piloting [P4, P7]; Supply Chain Disclosure [P5]</i>	Initial Simpl security MIM pilots in 3 EU Data Spaces (Manufacturing, Health, Energy); CRA-baseline SBOM tooling deployed by early adopters in Open RAN supply chains; CAMARA API harmonisation tested in 2–3 bilateral 5G corridor agreements; automated compliance negotiator prototypes validated in controlled IPCEI-CIS pilot environments; semantic security ontology (STIX 2.1/TAXII) integration for federated SOC interoperability under active R&I; ZKP computational overhead benchmarking for supply chain attestation use cases confirming batch verification viability.	Automated Simpl MIM policy translation operational across $\geq 50\%$ of active EU Data Spaces; Runtime SBOMs with cryptographic proof of execution verifying containerised components dynamically across multi-provider continuum nodes; ZKP benchmarks achieving batch verification viability for supply chain attestation; CAMARA-based cross-border edge compute roaming active across 5+ EU operator pairs; Data Visa mechanisms enabling AI model migration without manual legal review; inter-operator east-west interface standards ratified and operational in IPCEI-CIS corridor deployments. Advanced applied R&D: next-generation semantic interoperability for post-quantum supply chains; federated digital twin infrastructure (CL3.5) for pre-production security validation.	Provider-agnostic EU trust fabric operational continent-wide with real-time automated security orchestration; ZKP-verified supply chain attestation eliminates unverified components from EU critical infrastructure at continental scale; unified EU NaaS platform legally operable as one data jurisdiction; EU supply chain governance frameworks adopted as international reference models; federated cyber range infrastructure (CL3.5) providing pre-production validation for all cross-border security configurations before continental deployment.
	<i>Transition criterion: at least one functional Simpl MIM pilot demonstrated in a named Data Space with automated cross-provider security policy enforcement verified by an independent audit.</i>	<i>Transition criterion: Runtime SBOM with cryptographic execution proof operational in at least one Open RAN deployment; CAMARA cross-border roaming active across 5+ operator pairs with documented SLA compliance.</i>	<i>End-state criterion: EU supply chain governance and NaaS federation frameworks referenced in at least two international standards or bilateral digital trade agreements.</i>

Continued...

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
(2) Policy, Regulatory & Standards <i>Status: CRA/NIS2/DORA enforcement creating supply chain and interop mandates</i>	CRA Annex I Part II SBOM disclosure obligations create the mandatory supply chain transparency baseline [ref]; NIS2 Article 21(2)(h) supply chain security measures mandatory for essential entities [ref]; DORA Article 28 ICT third-party risk management requirements active for financial sector [ref]; Data Act interoperability obligations entering into force; Simpl-Open official launch with initial security MIM definitions establishing the technical baseline for CEN/CENELEC standardisation; GSMA CAMARA API standards reaching v1.x publication milestone. <i>Critical policy gap: Data Act implementing acts specifying interoperability technical requirements for security interfaces must be finalised within Phase 1 to provide standards basis; delayed publication creates a compliance gap for cross-border Data Space deployments.</i>	Security MIMs adopted as CEN/CENELEC standards with mandatory reference in EU Cloud Rulebook — the single binding gate condition for non-voluntary hyperscaler compliance; DORA Article 28 continuous third-party monitoring requirements creating procurement pull for runtime SBOM verification tooling; Data Act full enforcement driving interoperability adoption beyond early adopters; EU NaaS federation legal framework (Data Visa mechanisms) in active legislative development; EUCS automated certification operational for Simpl-federated cloud services [ref]. <i>Critical policy gate: security MIMs adopted as CEN/CENELEC standards with mandatory Cloud Rulebook reference — this is the single gate condition controlling the P4 and P7 SSL transition from Emerging EU Ecosystem to Full Sovereign Autonomy.</i>	“Schengen for Cloud” legal framework enacted, resolving US CLOUD Act extra-territorial jurisdiction conflict and enabling EU NaaS to operate as a single data jurisdiction; EU supply chain governance model referenced internationally; EUCS automated certification adopted in bilateral trade agreements as mutual recognition framework; EU Cloud Rulebook MIM requirements referenced in ISO/IEC service interoperability standards; ZKP supply chain verification standard adopted as international reference. <i>End-state governance criterion: Schengen for Cloud enacted; at least two bilateral mutual recognition agreements referencing EU supply chain certification; EU interoperability standards referenced in ISO/IEC.</i>
(3) Operational Deployment & Adoption <i>Deployment: Hyperscaler API dominance; bilateral pilots only</i>	Static CRA-baseline SBOM declarations submitted as point-in-time documents without runtime verification; CAMARA APIs tested in 2–3 bilateral 5G corridor agreements between individual EU operators; proprietary hyperscaler APIs (AWS, Azure, GCP) dominate all cross-provider integration; Simpl security MIM pilots deployed in proof-of-concept contexts in Manufacturing and Health Data Spaces; supply chain visibility confined to tier-1 vendor declarations without cryptographic verification; no operational pan-EU NaaS platform. Deployment actors: EU-funded research consortia and telecom operators in designated corridor pilots; SME participation absent due to compliance cost. <i>Transition criterion: operational Simpl MIM deployment in at least two EU Data Spaces verified by independent audit; at least three EU telecom operators in bilateral CAMARA-based cross-border roaming arrangement.</i>	Automated Simpl MIM policy translation operational across $\geq 50\%$ of active EU Data Spaces with cryptographically verifiable enforcement backed by Cluster 1 TEE attestation; Runtime SBOMs with execution proofs deployed in Open RAN supply chains of 3+ EU operators; CAMARA-standardised cross-border edge compute roaming active across 5+ EU operator pairs; supply chain quarantine (eBPF-isolated slices) operational for unverified components; SME-accessible SBOM and MIM compliance tooling available via Digital Europe Programme support schemes. <i>Transition criterion: deployment across 5+ Member States or 3+ critical infrastructure sectors; documented SME participation; supply chain quarantine operational in at least one production environment.</i>	All EU Common Data Spaces operating on Simpl-based security interoperability with automated MIM compliance verification; ZKP-verified supply chain attestation integrated into all EU critical infrastructure procurement workflows without human sign-off; unified NaaS platform enabling EU operators to compete at hyperscaler scale within the EU regulated market; supply chain quarantine automated with zero-human intervention; federated cyber range (CL3.5) providing pre-production validation for cross-border configurations before continental deployment. <i>End-state deployment criterion: security interoperability is the unconditional default; no EU critical infrastructure procurement proceeds without Simpl MIM compliance verification; supply chain clarity is structurally guaranteed, not audited.</i>

Continued. . .

Analytical Dimension	Phase 1 (2025–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
(4) Market Uptake & Strategic Sovereignty SSL: Emerging EU Ecosystem [P4] / Hyperscaler Dominance [P7]	AWS, Azure, and GCP collectively control approximately 65% of EU cloud market with proprietary API lock-in; no operational pan-EU NaaS platform; SBOM ecosystem fragmented between US-tooling vendors (Anchore, Snyk) and immature EU alternatives; SME participation constrained by compliance cost for CRA-mandated supply chain verification tooling [ref]; US CLOUD Act extra-territorial jurisdiction remains a structural sovereignty risk for all EU data processed on non-EU-domiciled infrastructure. SSL baseline: Emerging EU Ecosystem (P4); Hyperscaler Platform Dominance (P7); High reliance on fragmented non-EU supply chain auditing (P5). Market transition criterion: at least one EU MIM-compliant cloud service achieves a verifiable procurement award in a regulated-sector Data Space.	SSL: Growing Pan-EU Sovereignty EU-based cloud providers holding measurably growing share in regulated-sector procurement via Simpl MIM compliance; hyperscaler API lock-in reduced to legacy deployments as MIM-compliant alternatives gain procurement weight in 5+ Member States; EU NaaS consortium beginning to compete with US hyperscaler scale in defined vertical sectors; supply chain dependency on non-EU tier-1 vendors reduced in Open RAN deployments; ZKP-verified supply chain attestation enabling EU actors to demonstrate compliance superiority over foreign self-certification declarations. Market transition criterion: EU MIM-compliant cloud services hold at least 30% of new regulated-sector Data Space procurement; measurable reduction in hyperscaler API lock-in for new critical infrastructure deployments.	SSL: Full Federated Sovereign Autonomy EU cloud and edge providers achieve hyperscaler-comparable scale within EU regulated market through NaaS federation; US CLOUD Act jurisdiction exposure eliminated for EU data processed on federated sovereign infrastructure; supply chain transparency sufficient to exclude non-compliant components without bilateral negotiation; EU interoperability and supply chain standards exported to and adopted by partner jurisdictions; structural foreign dependency in cloud, edge, and supply chain domains reduced to managed residual legacy under active replacement programmes. End-state SSL criterion: EU NaaS achieves hyperscaler-comparable scale in EU regulated market; Schengen for Cloud eliminates CLOUD Act exposure for new deployments; EU is net exporter of supply chain governance standards.
Critical Enabler Gate Phase transition:	Simpl-Open official launch with security MIM definitions published as open standards [ref] establishing the technical baseline for cross-provider security policy automation and providing the specification reference that makes CEN/CENELEC standardisation in Phase 2 possible.	Security MIMs adopted as CEN/CENELEC standards with mandatory reference in EU Cloud Rulebook creating the regulatory basis for non-voluntary hyperscaler compliance and transforming voluntary interoperability into a structural market requirement enforceable against all cloud service providers operating in the EU market.	“Schengen for Cloud” legal framework enacted resolving US CLOUD Act extra-territorial jurisdiction conflict and enabling the EU NaaS platform to operate legally as a single data jurisdiction, achieving the commercial scale conditions for Full Federated Sovereign Autonomy.
Sub-Priority ODP / SSL Decomposition — individual maturity tracking per constituent priority across phases.			
P4: Security Interoperability via Simpl Middleware	ODP: Fragmented Piloting SSL: Emerging EU Ecosystem	ODP: Standardised Integration SSL: Full Sovereign Autonomy	ODP: Ubiquitous Operation SSL: Full Sovereign Autonomy
P5: Supply Chain Verification & Secured Infrastructure	ODP: Full Supply Chain Disclosure SSL: High Reliance on Fragmented Non-EU Auditing	ODP: Dynamic Runtime Verification SSL: Emerging EU Supply Chain Tracking Frameworks	ODP: ZKP-Verified Attestation & Full Restriction SSL: Full Sovereign Supply Chain Control
P7: Pan-European NaaS Federation	ODP: Bilateral Interconnection SSL: Hyperscaler Platform Dominance	ODP: Federated Edge Roaming SSL: Pan-EU Consortium Competes	ODP: True Digital Single Market SSL: European Scale Parity Achieved